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KONGU ENGINEERING COLLEGE
PERUNDURAI
LOVE * KNOWLEDGE * SERVICE
Estd : 1984

22CSP72 – PROJECT WORK II
PHASE I REVIEW

Evidence-Based Fake News Detection with Calibrated Scoring

Date : 10-10-2026

Team Number : CO5

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PROBLEM STATEMENT & OBJECTIVES

Fake news detectors judge an article by its wording, its headline and whether it names sources, and the writer controls all of these. When we tested our earlier pipeline with invented stories in calm newsroom style, they averaged 7.3 out of 10 and four of six deceptive items read as credible.

01

Evidence Over Style

Decide a story by what independent sources report, not by how it is written.

02

Independent Sources

Shared owners, wire copies and reprints count once: five copies are one source.

03

Catch False Premises

Check named people against Wikipedia to expose an invented role or office.

04

Verify the Publisher

Identify the outlet from its web address, so a copycat site cannot borrow a name.

05

Calibrated Verdict

Measured likelihood ratios give a probability of fabrication that matches outcomes.

06

Honest Abstention

Refuse input with nothing to check; report a failed search as "could not verify".

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LITERATURE SURVEY

Reference	What They Did	What's Missing
Braun et al., ICML 2025 (DEFAME, base paper)	Multimodal fact-checking agent that plans web and image searches and writes a justified verdict	Does not weigh source credibility; no probability
Verma, arXiv 2026	Audited the ISOT corpus: the subject field alone gives F1 = 1.000	High benchmark accuracy reflects corpus artefacts
Chen & Shu, ICLR 2024	Showed LLM-written misinformation is harder to detect	Style-based detectors miss fluent fakes
Hu et al., AAAI 2024	LLM rationales guide a small fake news classifier	Still judges the article text alone
Wang et al., ACM Web 2024	LLM weighs competing crowd reports about a claim	Needs many reports for each claim
Schlichtkrull et al., NeurIPS 2023	AVeriTeC: real-world claims paired with web evidence	A benchmark, not a reader-facing tool
Lin et al., PNAS Nexus 2023	Combined six rating sets into one score for 11,520 news domains	Rates outlets, not individual stories

BASE PAPER: DEFAME vs PROPOSED SYSTEM

Braun et al., *"DEFAME: Dynamic Evidence-based FAct-checking with Multimodal Experts"*, ICML 2025

Aspect	DEFAME (ICML 2025)	Proposed System
Evidence	Web search, image search, reverse image search, geolocation	News index, Wikipedia entries, publisher record, domain registration
Source credibility	Not weighed; relies on the search engine	Publisher identified by web address; 90 curated and 11,520 rated domains
Independence	Not considered	Shared owners, wire copies and reprints merged
Output	One of four labels with a written justification	Verdict, calibrated probability, the rule that fired and quoted grounds
Missing evidence	"Not enough information"; re-plans up to three times	Refuses uncheckable input; CANNOT VERIFY with the reason
Failed check	Not separated from "no evidence"	Reported as "verification unavailable", never as absence
Model and cost	GPT-4o; about 4 min and 13 US cents per claim	9 of 10 model tasks on a local 8B model; 22 to 112 s per article

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RESEARCH GAP

Gap in existing work	How the proposed system closes it
Writing signals the author controls can make a fake look credible	Style may only count against a story, never for it
Retrieved articles are counted, so syndicated copies inflate support	Shared owners, wire copies and reprints count as one source
The outlet is taken from the name the page gives itself	Publisher read from the web address; lookalike sites flagged
A false premise describes an event nobody reports, so search finds nothing	Named people checked against their Wikipedia entries
Hand-picked weights give scores that are not probabilities	Likelihood ratios measured on labelled data, then calibrated
A failed or empty search is shown as evidence against the story	Failed search reported as unavailable; the system abstains

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SYSTEM ARCHITECTURE

Client: React 19 single-page application

- Front page with trust stamps
- Article page
- Check Anything
- Compare Coverage
- Reader dashboard

API: Express 5 on Node.js 24

- Accounts (JWT, bcrypt)
- Access control (JWT or key)
- Link guard (no private addresses)
- NDJSON progress stream
- Report cache (14 days, complete only)

Analysis agents

- Ingestion: page reader, OCR, captions
- Admissibility gate
- Claim extraction
- Language detection
- Corroboration search
- Relevance gate
- Independence analysis
- Stance judgement
- Premise check (Wikipedia)
- Publisher identity and domain age
- Writing checks (4)
- Weight of evidence and verdict rules

Language-model gateway: task routing, retry and back-off, JSON repair

- Ollama, Llama 3.1 8B (9 tasks)
- NVIDIA NIM (premise check)
- Google Gemini (fallback)
- Rule-based fallback analysers

Data

- MongoDB: users, history, activity, report and summary cache
- Outlet ratings: 90 curated, 11,520 rated domains
- Ownership map
- Measured evidence weights and routing table

External services

- NewsAPI news index
- Wikipedia API
- RDAP domain registries
- YouTube captions
- Publisher web pages

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PROCESSING PIPELINE

Article link: publisher page | Pasted text | Screenshot: Tesseract OCR | YouTube video: caption track

Normalise
title, text, source, provenance; language

Admissibility gate
needs an anchor (name, place, number, date), 40+ characters, 8+ words

no → **Refused**
reader told what is missing

Claim extraction: up to two claims, 3 to 4 search terms each; English terms added for other languages

Corroboration

- Search the news index, own outlet excluded
- Relevance gate: must share a name, place or figure
- Merge shared owners, wire copies, reprints
- Stance: supports or contradicts

Premise check

- Resolve up to two named subjects
- Fetch the Wikipedia entry
- Admit a conflict only if: same subject, cannot both be true, story's own voice, quoted line

Publisher identity

- Outlet from the web address, never the page name
- Rating: 90 curated, else 11,520 rated domains
- Impersonation flag; RDAP domain age under 90 days

Writing checks

- Clickbait in the headline
- Slanted or one-sided language
- Persuasion techniques (SemEval)
- Transparency checklist; each quotes its span

Weight of evidence: prior from the publisher band → each observation as a likelihood ratio in decibans → writing checks clamped at max(0, W) → correlated checks damped × 0.379, capped at 8 dB → isotonic calibration → clamp applied again

Verdict rules R0 to R9
ordered; the first match decides and is named

Report: verdict, probability of fabrication, grounds and quoted text; streamed as each check ends; cached only when every check ran

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WEIGHT-OF-EVIDENCE SCORING

Each observation becomes a likelihood ratio, in decibans

$\Lambda(o) = P(o \mid \text{fabricated}) / P(o \mid \text{genuine})$
$W(o) = 10 \log_{10} \Lambda(o)$

Evidence adds up in log-odds

$\text{logit}\, P(\text{fabricated} \mid O) = \text{logit}\, \pi + \Sigma\, W(o)$

One-sided rule for what the author controls

$W\dagger(o) = \max(0, W(o))$

Correlated writing checks are damped

mean $r = 0.55$ over 4 checks $\rightarrow$ 1.52 effective checks
summed weight $\times$ 0.379, capped at 8 dB

Prior by publisher band

Publisher band	Prior
Strong factual record	0.02
Ordinary record	0.06
Unrated outlet	0.15
Poor record or impersonating	0.25
Social media post	0.35

Ratios estimated on 119 labelled ISOT articles with a Jeffreys prior and shrunk to the cautious end of a 95% interval. Isotonic calibration fitted on held-out data, then the one-sided rule is applied again. Priors are declared, not measured.

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VERDICT ENGINE

Rule	Condition (first match decides)	Verdict shown
R0	The publisher is a known satirical outlet	SATIRE
R1	Two or more independent outlets contradict a claim	FAKE (false)
R2	One independent outlet contradicts a claim	FAKE (likely false)
R3	A reference entry contradicts the story's premise	FAKE (likely false)
R4	The search could not run, or the story is newer than the news index	CANNOT VERIFY
R5	No checkable claim is left after extraction	NOT A FACTUAL CLAIM
R6	Two or more independent outlets support the claims	REAL (confirmed)
R7	One independent outlet supports the claims	REAL (likely true)
R8	High-impact claim, uncorroborated, three or more risk signals	FAKE (likely false)
R9	None of the above	CANNOT VERIFY

Input that fails the admissibility gate is refused before these rules run. Each verdict also caps the score shown, so an unconfirmed story can never read as trustworthy.

MODULES OF THE PROJECT

Built by Review 2

- News platform: feed, login, history, dashboard
- Clickbait, bias, persuasion and transparency checks
- Cross-source claim verification
- Multi-provider AI gateway with failover
- Compare Coverage page
- Analyse any article by link
- Rule-based fallback analysers

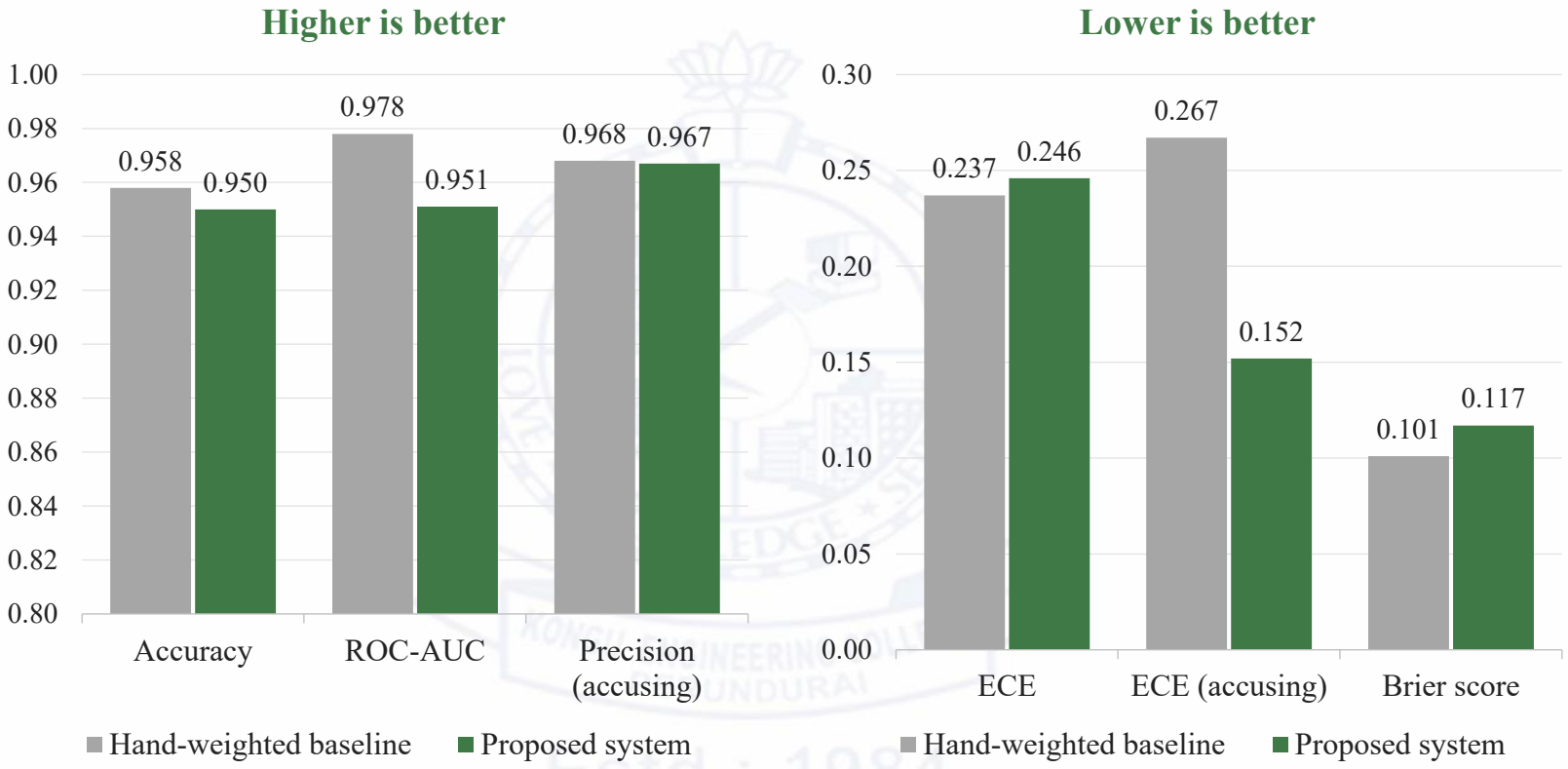
Added for Review 3

- Evidence-first verdict engine (R0 to R9)
- Weight-of-evidence scoring with calibration
- Source independence analysis
- Premise check against Wikipedia
- Publisher identity from the web address, 11,520 rated domains, domain age
- Check Anything: text, screenshot, YouTube, Tamil
- Admissibility gate for uncheckable input
- Local model routing: 9 of 10 tasks on Ollama
- API access control and link guard
- 182 backend and 14 frontend tests

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RESULTS: HELD-OUT EVALUATION

120 held-out ISOT articles; source hidden, agency tags removed



Accuracy is the same to within one article **(114 vs 115 of 120)**. Calibration error when the system accuses fell from **0.267 to 0.152**, 43% lower. Turning the one-sided rule off raises ROC-AUC to 0.983 but accusing ECE to 0.191.

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ADVERSARIAL & PUBLISHER TESTS

0 of 6

deceptive items rated credible
baseline: 4 of 6

6 of 6

verdicts as expected
0 of 2 genuine reports condemned

2% → 33%

fabricated story on a copycat BBC site
probability of fabrication

Lookalike publishers: identity from the page name vs from the web address

Address	By page name	By web address
bbc-breaking-news.xyz, calling itself "BBC News"	BBC News, 9/10	No rating; impersonation flagged
bbc-world.com	BBC News, 9/10	No rating
bbc.news-today.com	BBC News, 9/10	No rating
reuters-updates.info, calling itself "Reuters"	Reuters, 9.5/10	No rating; impersonation flagged
YouTube channel named "Associated Press"	AP, 9.5/10	No rating

END-TO-END TESTING

No.	Input	Expected	Result	Decided by	Time
1	Pasted text	REAL	REAL	5 independent outlets	100 s
2	News link	REAL	REAL	3 independent outlets	33 s
3	Screenshot	REAL	REAL	5 independent outlets	112 s
4	Tamil text	REAL	REAL	4 independent outlets	30 s
5	YouTube video	REAL or CANNOT VERIFY	CANNOT VERIFY	No coverage found	55 s
6	Pasted text	FAKE	FAKE	Premise contradicted	22 s
7	Screenshot	FAKE	FAKE	Premise contradicted	30 s
8	Pasted text	CANNOT VERIFY	CANNOT VERIFY	No coverage found	25 s
9	Pasted text	CANNOT VERIFY	CANNOT VERIFY	No coverage found	28 s
10	Pasted text	Refused	Refused	A question	< 1 s
11	Screenshot	Refused	Refused	A call to action	< 1 s
12	News link	Refused	Refused	Site blocks reading	< 1 s

11 of 12 matched exactly. The AP video, mostly a French interview, found no matching coverage on that run and returned CANNOT VERIFY; it was not called false.

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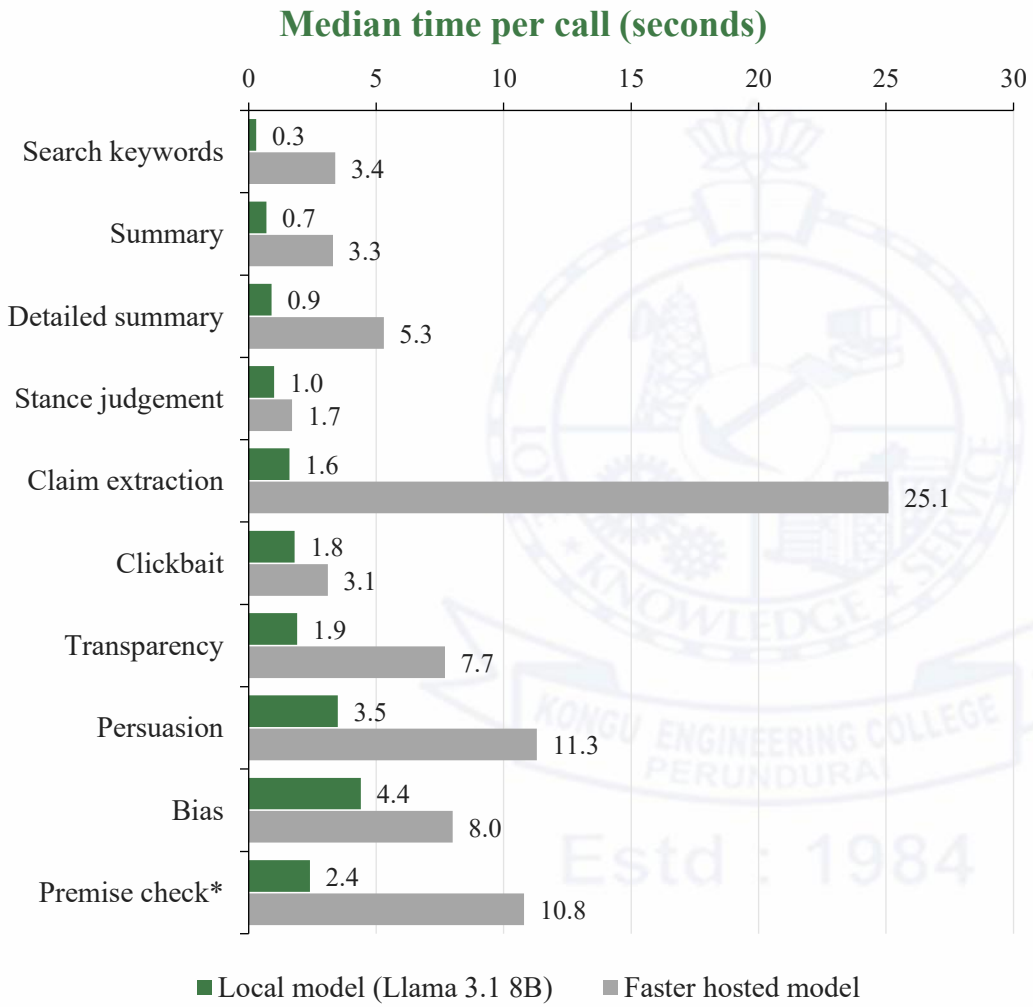
RESULTS: FACT-CHECKED RULINGS

Rulings	Claims	Desk's verdict	Acceptable	No verdict	Wrong
Development 1–120, final code	120	57	27	33	3
Sealed 121–320, first candidate	200	97	34	52	17
Sealed 121–320, final code	200	102	39	47	12
of which rated false	130	98	4	26	2
of which rated misleading	57	1	35	14	7
of which rated true	13	3	0	7	3

320 claims ruled on by 29 fact-checking desks; each desk's own ruling is hidden during a run. Rulings 121–320 are sealed: run whole, totals only, never read or tuned on. Final code: 141 of 200 desk's or acceptable verdict, 12 wrong. First candidate scored on labels later corrected (4 true → false).

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PERFORMANCE & MODEL ROUTING



** Premise check moved off the local model after it missed the known contradiction in 3 of 3 runs; it is now led by Gemini. The deployed server has no GPU, so there these tasks start at NIM.*

8 of 16

model tasks run on a local GPU when one is present

1.47 s

until the first check is on screen

2.7 ms

to return a cached report

519 + 42

backend and frontend tests passing

GIT TRACKING & DOCUMENTATION

Repo URL: github.com/Arumuganandhini/FakeNews_Detection

Repository

- main: stable project
- project: development branch
- API keys, .env files and datasets excluded
- Every change committed and pushed
- CI reruns all tests and builds on every push

Documentation

- Project report in the college format: 6 chapters, appendix, references, SDG page (73 pages)
- docs/measurements.md: every result with the command that reproduces it
- backend/eval: held-out, adversarial, end-to-end and fact-checked rulings suites, with the sealed-test log
- Automated tests: 519 backend, 42 frontend, all passing

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SDG GOALS ADDRESSED

Reliable verification that any reader can check, without removing or blocking content.

SDG 16

Peace, Justice and Strong Institutions

Target 16.10, public access to information: every verdict comes with its reason, the outlets involved and quoted text. Impersonating pages are flagged; opinion is not judged.

SDG 4

Quality Education

Quoted evidence shows readers how to judge a story themselves, building media literacy instead of asking for blind trust.

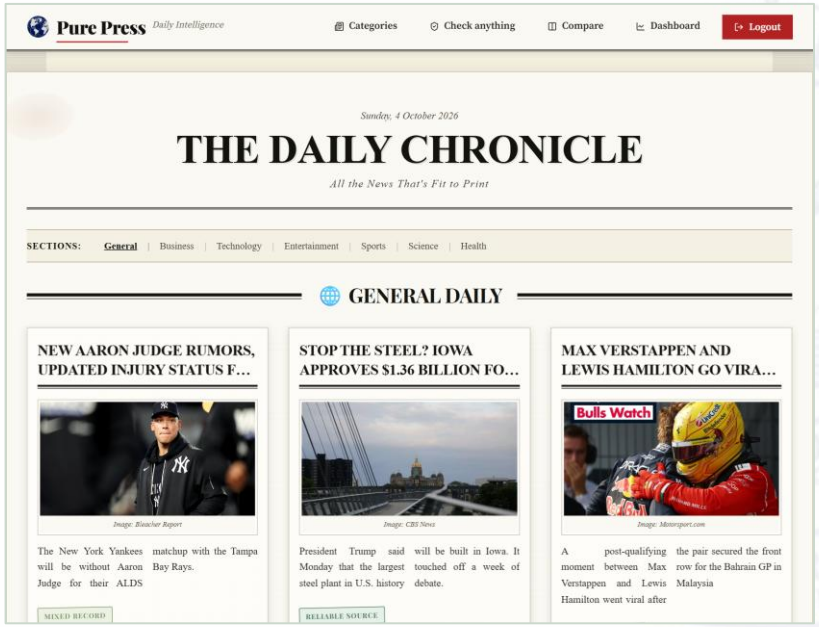
SDG 9

Industry, Innovation and Infrastructure

Verification at low cost: nine of ten model tasks run on a laptop GPU, and Tamil and other Indian-language input is accepted.

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OUTPUT



Front page with trust stamps



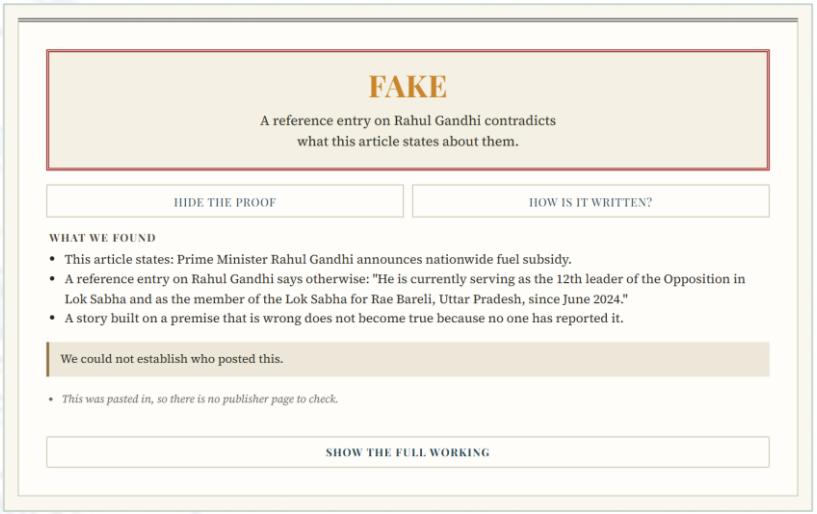
Check Anything page

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OUTPUT



Verdict: genuine report confirmed

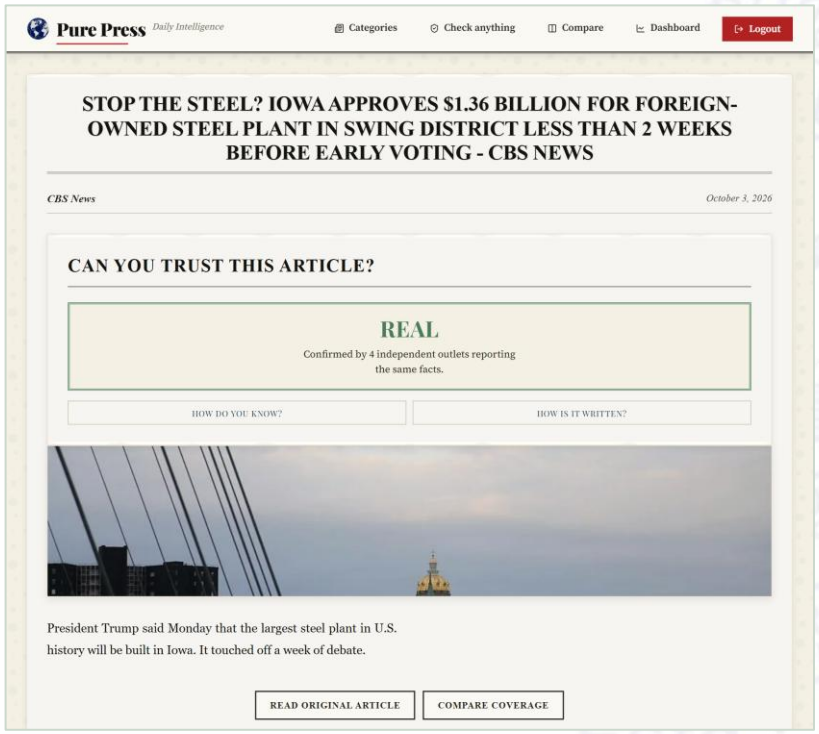


Verdict: false premise refuted

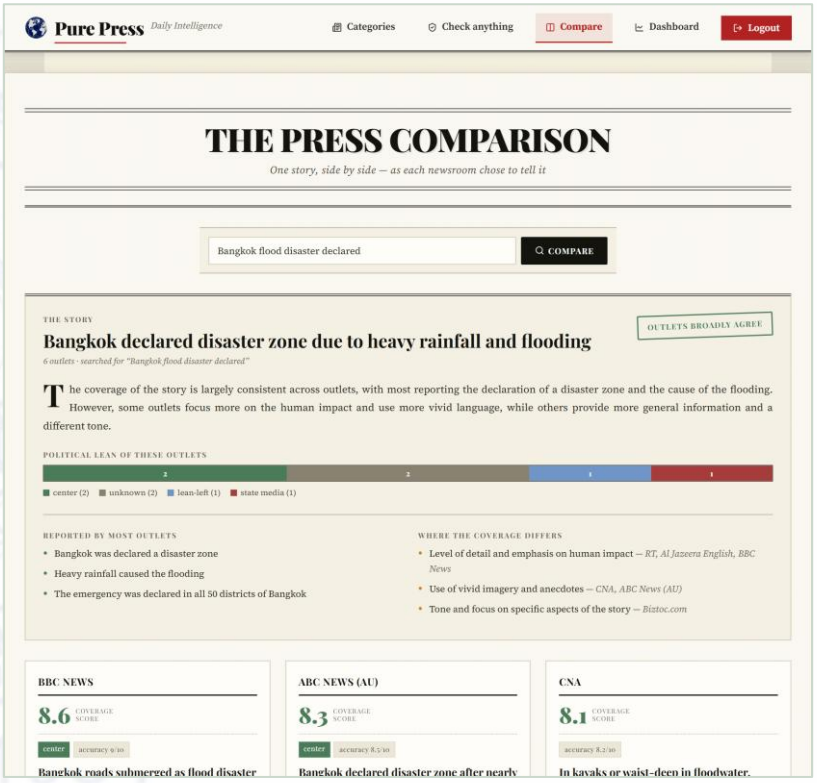
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OUTPUT



Article page



Coverage comparison

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CONCLUSION, FUTURE WORK & THANK YOU

Conclusion

A story is judged by independent evidence, not by its writing. On 120 held-out articles the system matched the baseline's accuracy and cut calibration error on accusations by 43%; it rated none of six careful fabrications as credible. On 200 sealed fact-checked claims it gave the desk's or an acceptable verdict for 141 and a wrong one for 12.

Future Work

Answer more of the claims left without a verdict (47 of 200) and of true claims (3 of 13 confirmed) by reading article bodies, not headlines; grow the labelled corpus beyond 239 articles; collect ratings for regional Indian outlets; run the judgements on our own models.

THANK YOU

A live demo is available for the panel.